EPIXA – Epilepsy eXpected Alert

Smart System for Early Prediction of Epileptic Seizures

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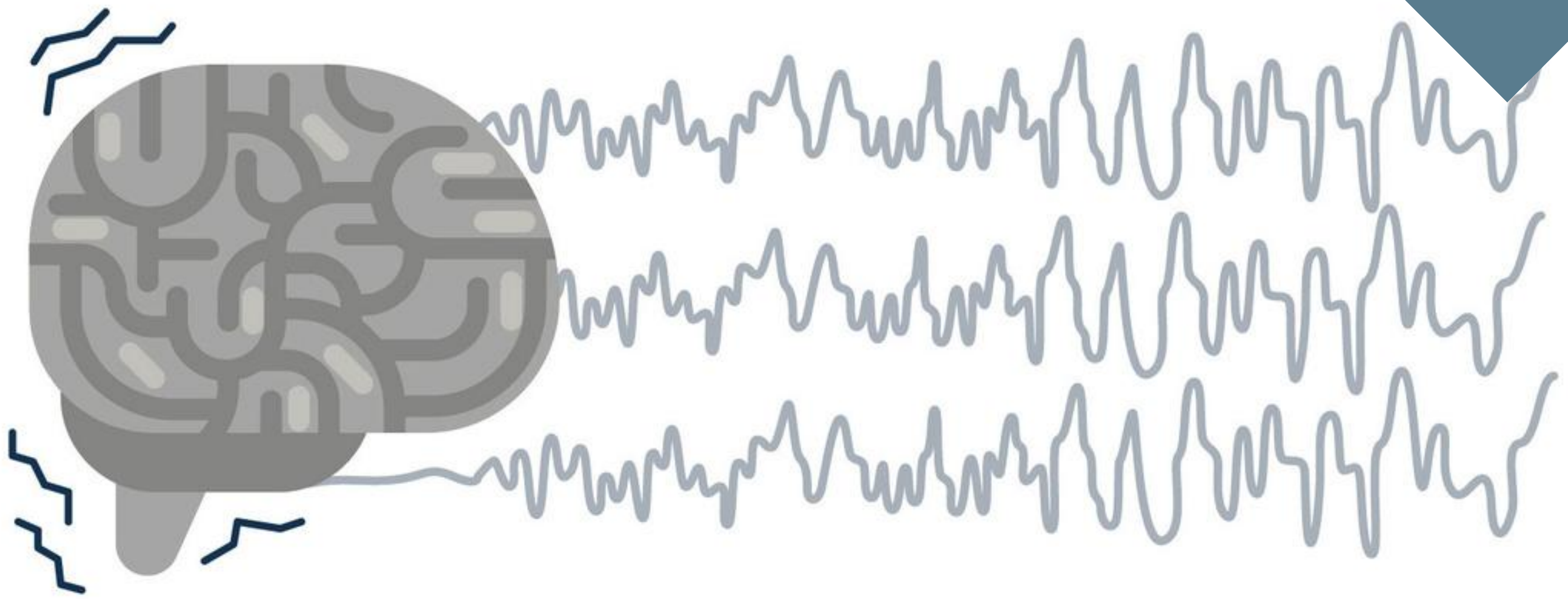
1. PROBLEM DEFINITION

1.1. EPILEPSY

- Over 50 million people worldwide suffer from epilepsy
- Up to 30% of patients cannot predict their seizures
- Sudden seizures may lead to serious injuries or death (SUDEP)
- Most current solutions only detect seizures after they occur

1.1. EPILEPSY



- A chronic neurological disorder.
- Characterized by recurrent, unpredictable seizures.

1.2. **Effects of Epilepsy**

- Sudden loss of consciousness.
- High risk of injuries and accidents.
- Constant fear and anxiety.

1.3. **Why Is It Dangerous?**

- Seizures can happen anytime, anywhere.
- Risk of SUDEP, especially during sleep.

1.4. Target Users & Impact

Patients (Epilepsy Patients)

- Live with unpredictable seizures and constant anxiety
- High risk of injury during seizure episodes
- Need real-time monitoring and early warnings

Caregivers (Family / Guardians)

- Continuous stress due to responsibility of monitoring
- Difficulty in providing 24/7 supervision
- Need instant alerts to respond fastly

Healthcare System

- Increased emergency admissions due to sudden seizures
- High pressure on hospitals and medical resources
- Need for preventive solutions to reduce emergency interventions

1. OUR SOLUTION

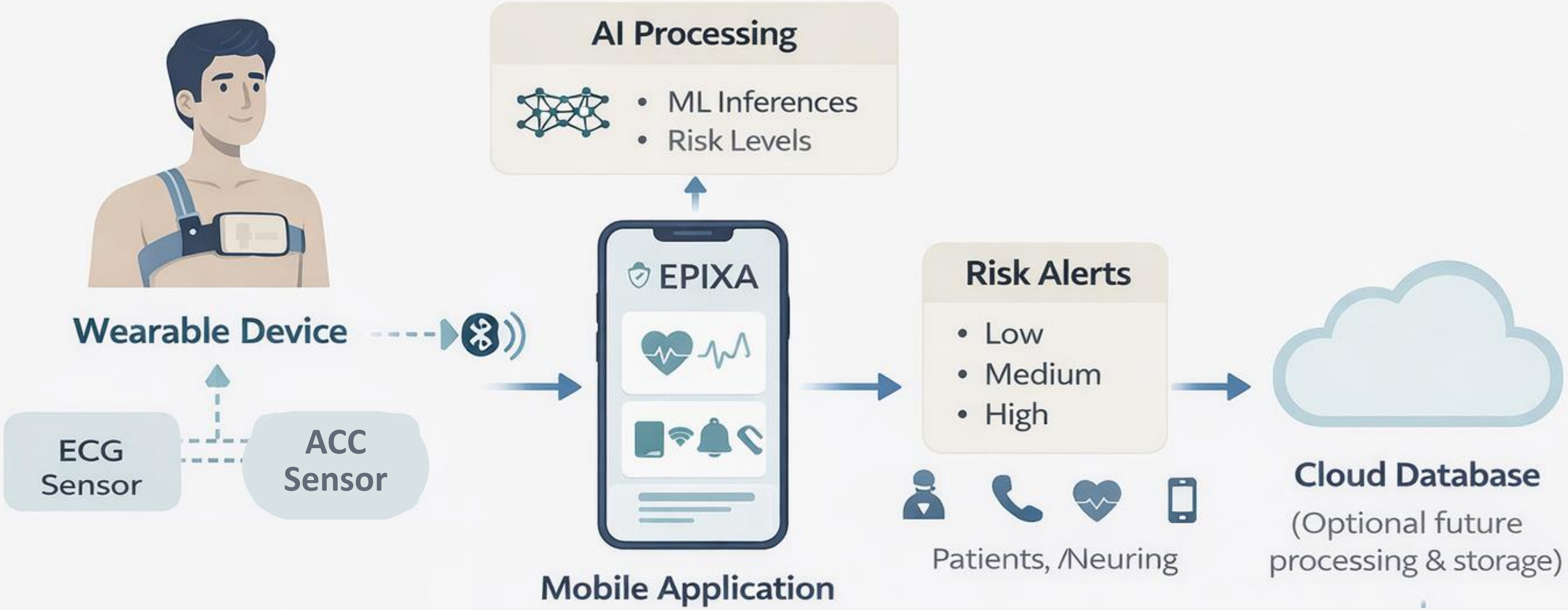
- Develop a **wearable, non-invasive** system for continuous seizure monitoring Without usage of **complex signals (eg. EEG signals)**.
- Enable **early seizure prediction** using AI and multimodal physiological signals.
- Provide **real-time alerts** to patients, caregivers, and clinicians
- Improve **patient safety, independence, and quality of life.**

1. OUR SOLUTION

Our Value Proposition

- We shift epilepsy care from reaction → prediction
- Predict seizures before they happen
- Provide real-time alerts for immediate action
- Enable continuous monitoring in daily life
- Deliver a non-invasive, wearable solution

3. SYSTEM ARCHITECTURE



The system focuses on early risk estimation rather than direct medical intervention.

How the System Works

- Sensors collect ECG & motion data
- AI model analyzes signals in real-time
- Detects pre-seizure patterns (pre-ictal phase)
- Sends instant alert to caregiver
- Enables fast response and reduces risk
- Simple. Real-time. Life-saving.

4.MARKET STUDY

MARKET STUDY

Device	Company	Technology	Detection	Prediction	Accuracy	Key Feature
Embrace2 /EpiMonitor	Empatica	EDA, ECG, Motion, Temp	✅ (Tonic-Clonic)	❌	85-90%	FDA Cleared, Low False Alarms
Smartwatches	Apple, Samsung	Heart Rate, Motion	⚠️ Limited (Abnormal Episodes)	❌	70-80%	Readily Available, High False Alarms
NightWatch	LivAssured	Heart Rate, Motion (Armband)	✅ (Nocturnal Tonic-Clonic)	❌	80-85%	Optimized for Night Use, CE Marked
Seer Sense	Seer Medical	cEEG (Behind-ear), ECG	✅ (Generalized)	❌	85-90%	Clinical Grade EEG, CE Marked
NeuroPace RNS System	NeuroPace	Intracranial EEG	✅ (Focal Seizures)	✅ (Prevents Seizures)	90% reduction	Only FDA-approved device for prediction & prevention
Research Prototypes	Universities / Labs	Multimodal + AI	✅	✅ (AI-based)	75-85%	In Development, Not Approved
BioSticker / BioStamp	BioIntelliSense	ECG, Respiration, Motion	⚠️ Limited (General Physiology)	❌	65-75%	FDA-cleared (General Use), Not Seizure-Specific

“NeuroPace RNS is an implantable clinical device and not comparable to wearable consumer solutions.”

MARKET GAP

Despite recent advances in wearable technologies:

- Most commercial devices are limited to seizure detection after onset.
- To date, no non-invasive wearable solution provides reliable early seizure prediction.

Furthermore, existing solutions lack full integration between:

- AI-based prediction.
- real-time mobile alerts.

Additionally, most solutions rely on either single-signal detection or invasive EEG systems.

COMPETITIVE ADVANTAGE

Feature	Ex
Seizure Detection	✅
Early Prediction	❌
Non-invasive	❌

BUSINESS & MARKET OPPORTUNITY

- Growing digital health & wearable market
- Increasing demand for remote patient monitoring

Target customers:

- Patients & families
- Hospitals & clinics
- Insurance providers

Potential for:

- Subscription-based mobile app
- Smart wearable integration
- Healthcare partnerships

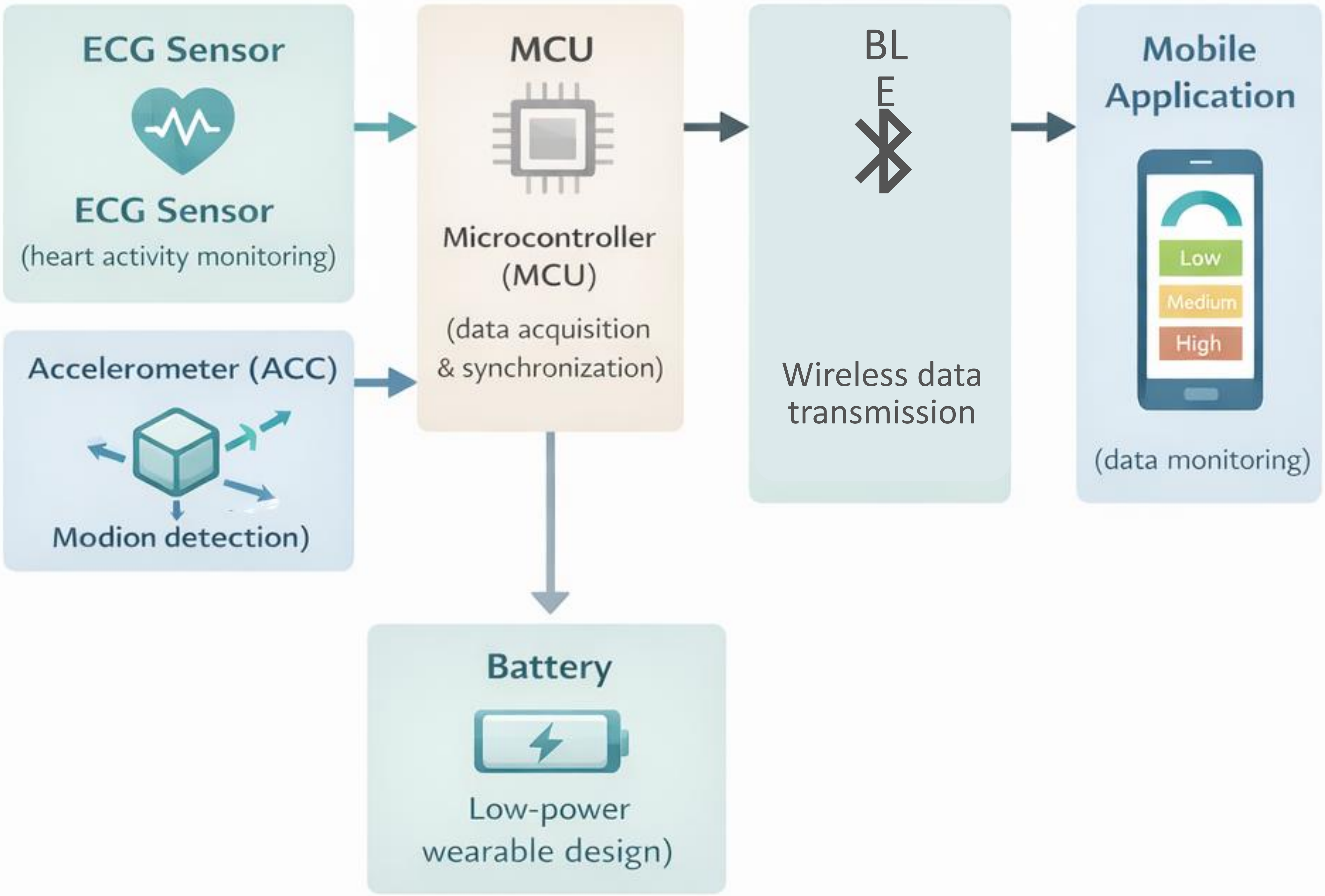
5.HARDWARE DEVICE

Design Constraints & System Requirements

- Non-invasive, daily-wear architecture
- Energy-efficient wearable design
- Low-latency, real-time risk estimation
- Personalizable AI for inter-patient variability

HARDWARE DEVICE

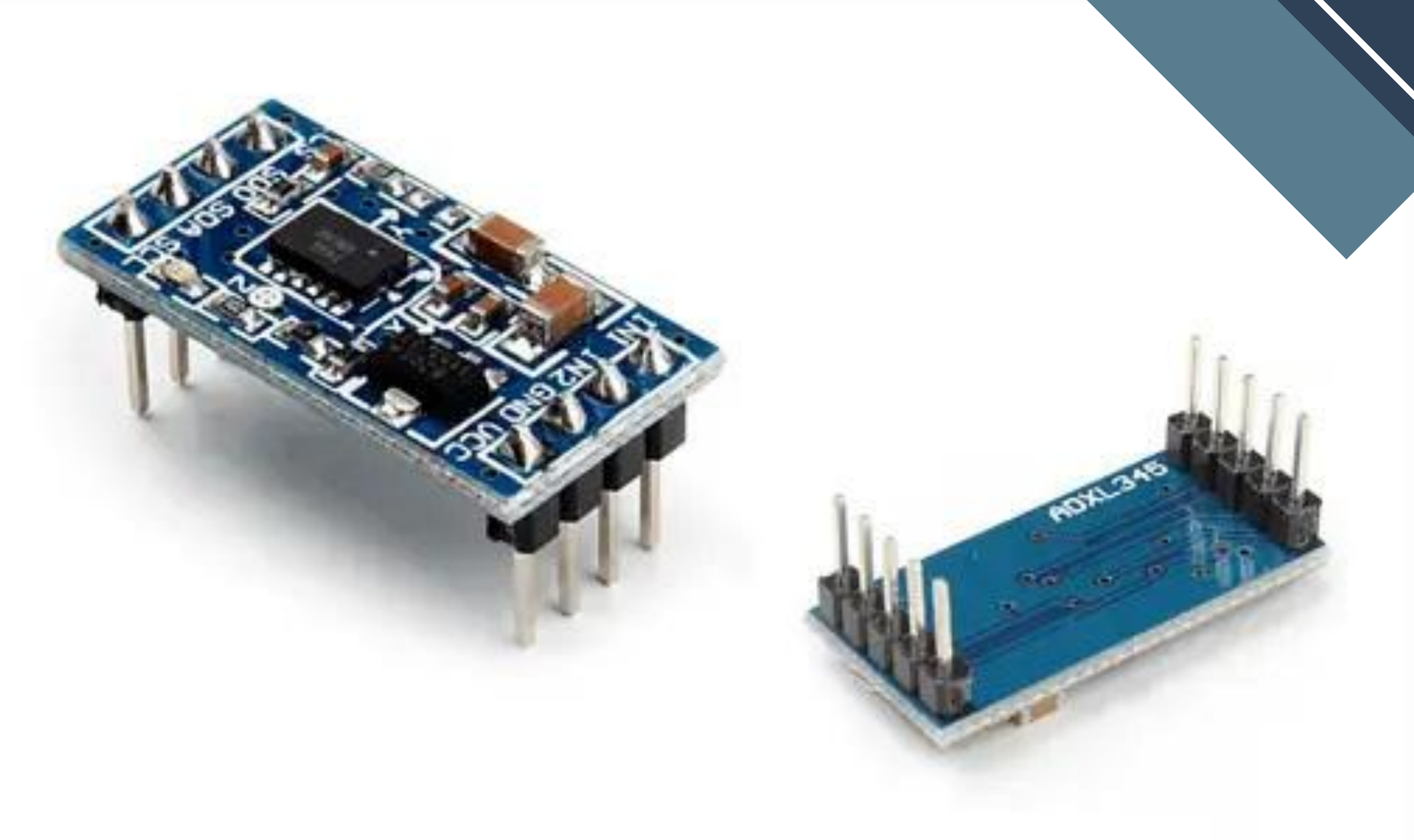
Circuit Diagram

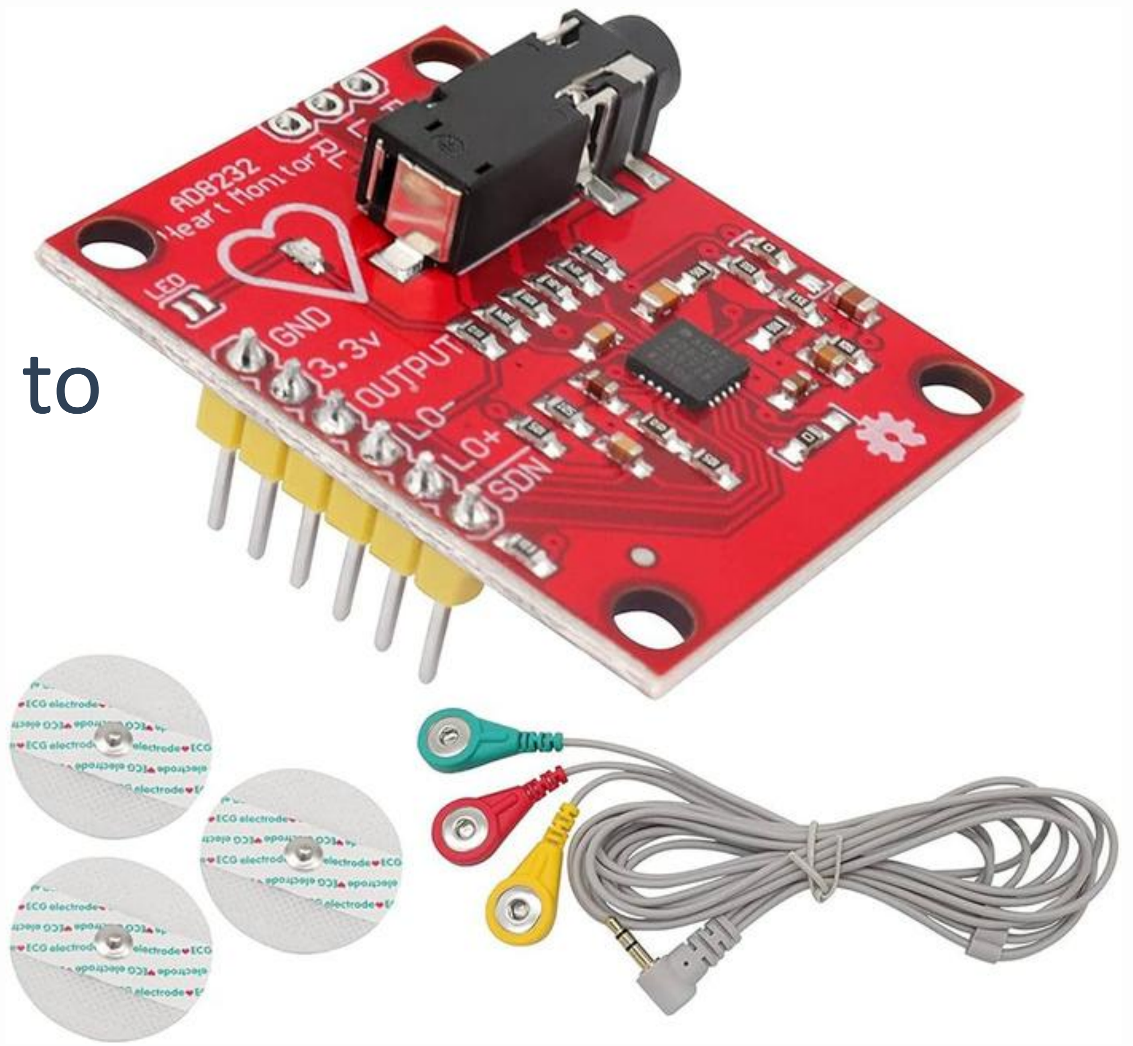


HARDWARE DEVICE

Components

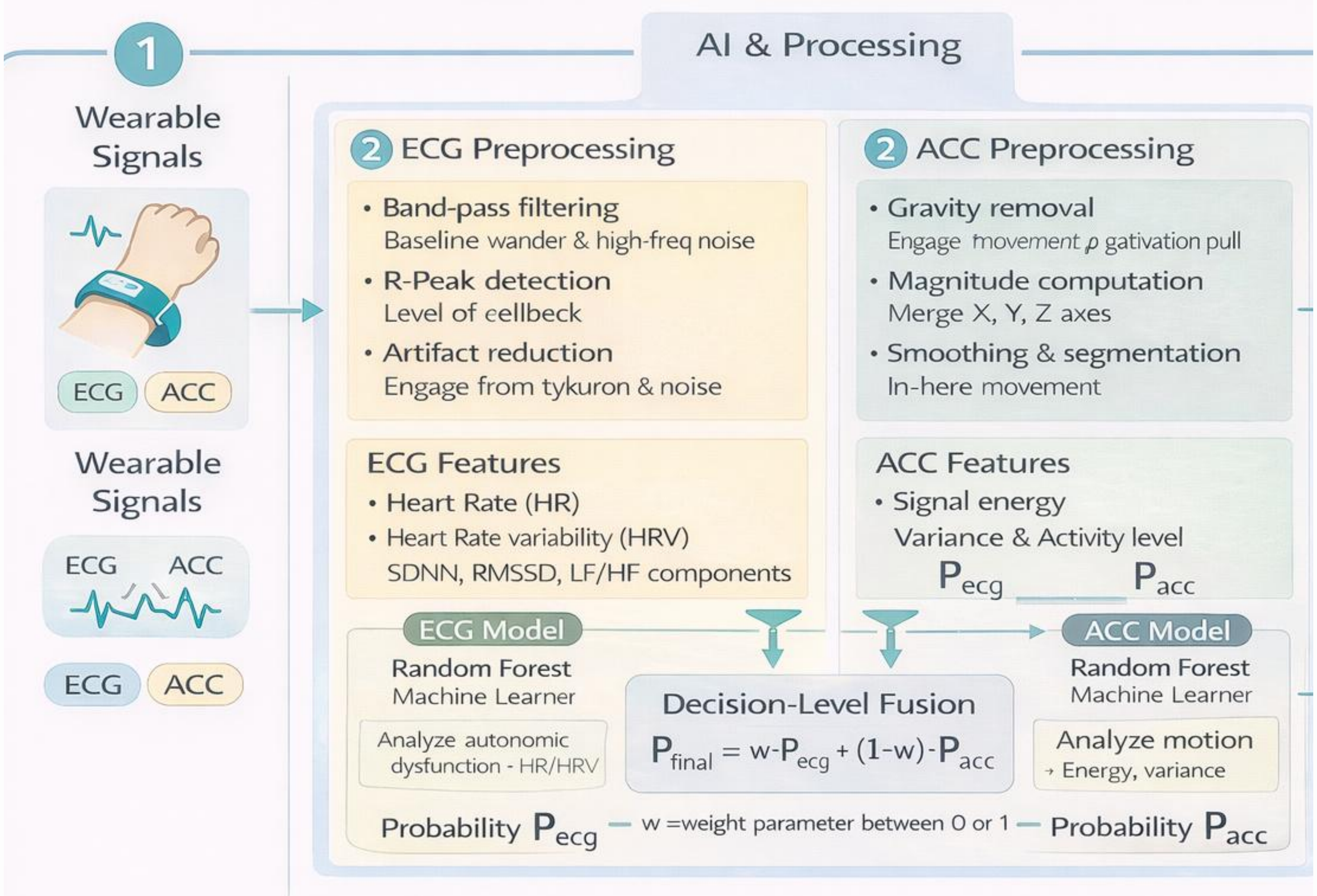
- ESP32 microcontroller => to collect the signals and send them via Bluetooth to mobile application.
- ECG sensor (AD8232) => three electrodes connected to chest.
- ACC sensor (ADXL345) => attached on chest to detect body movement.
- Power source 5-7 volt





6. PREDICTION MODEL

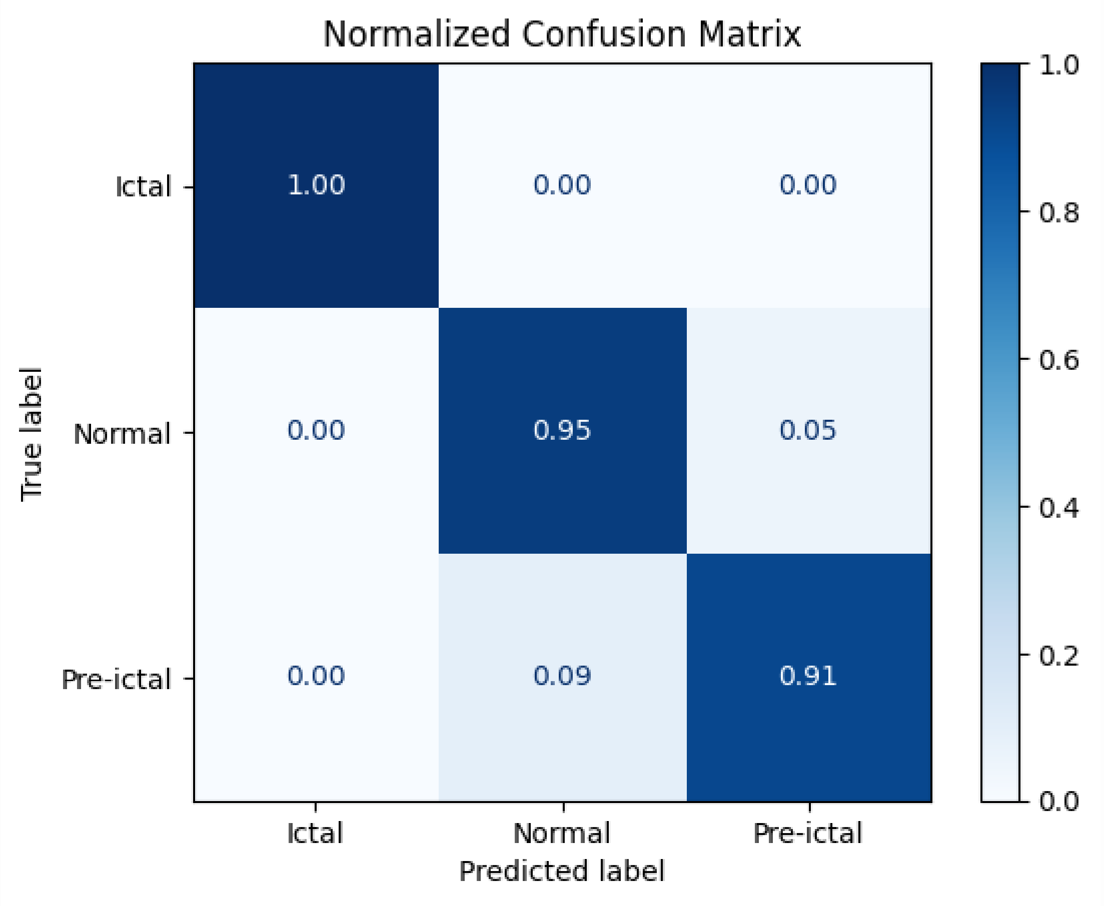
.Architecture



Model evaluation

The system currently performs best in patient-specific training settings, which is consistent with epilepsy literature.

1



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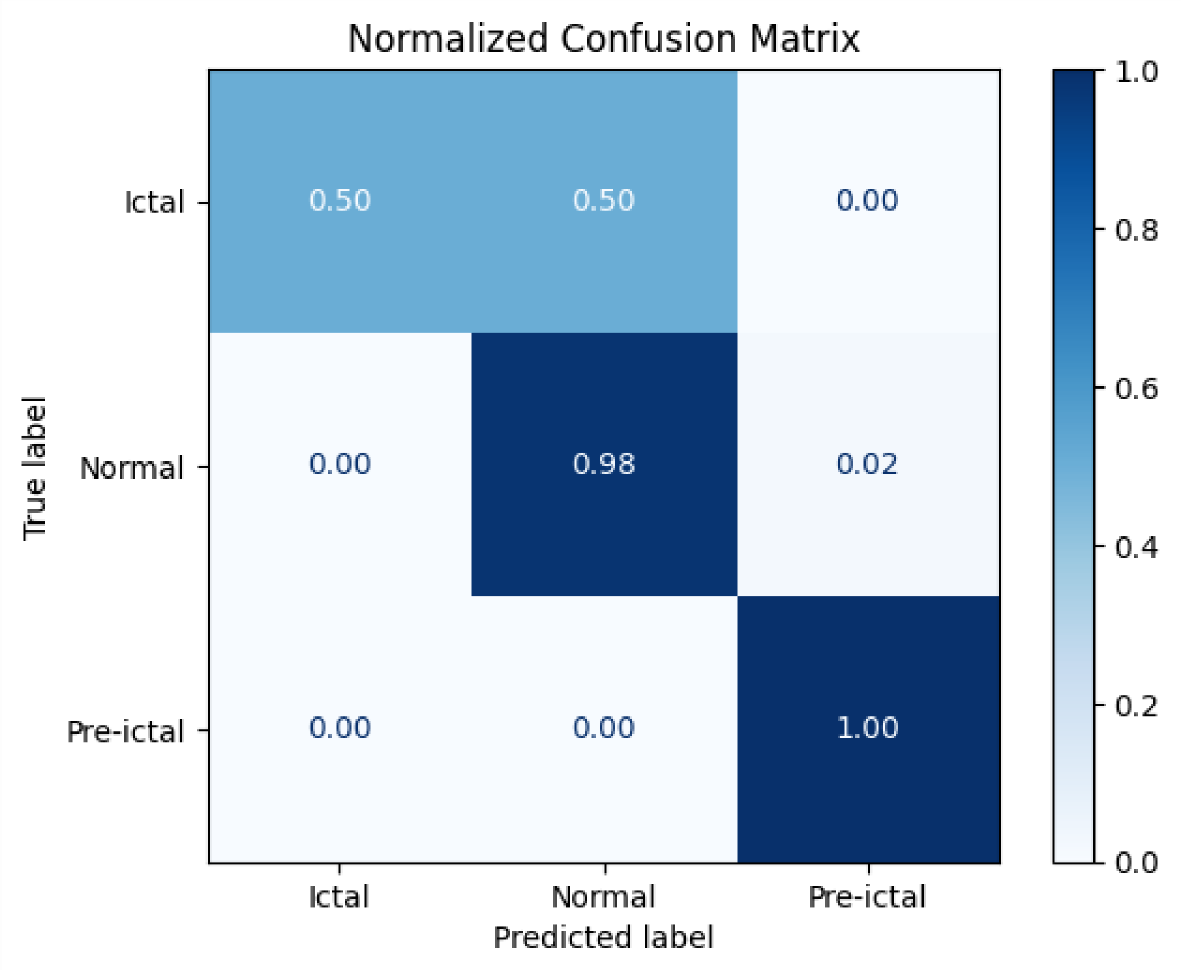

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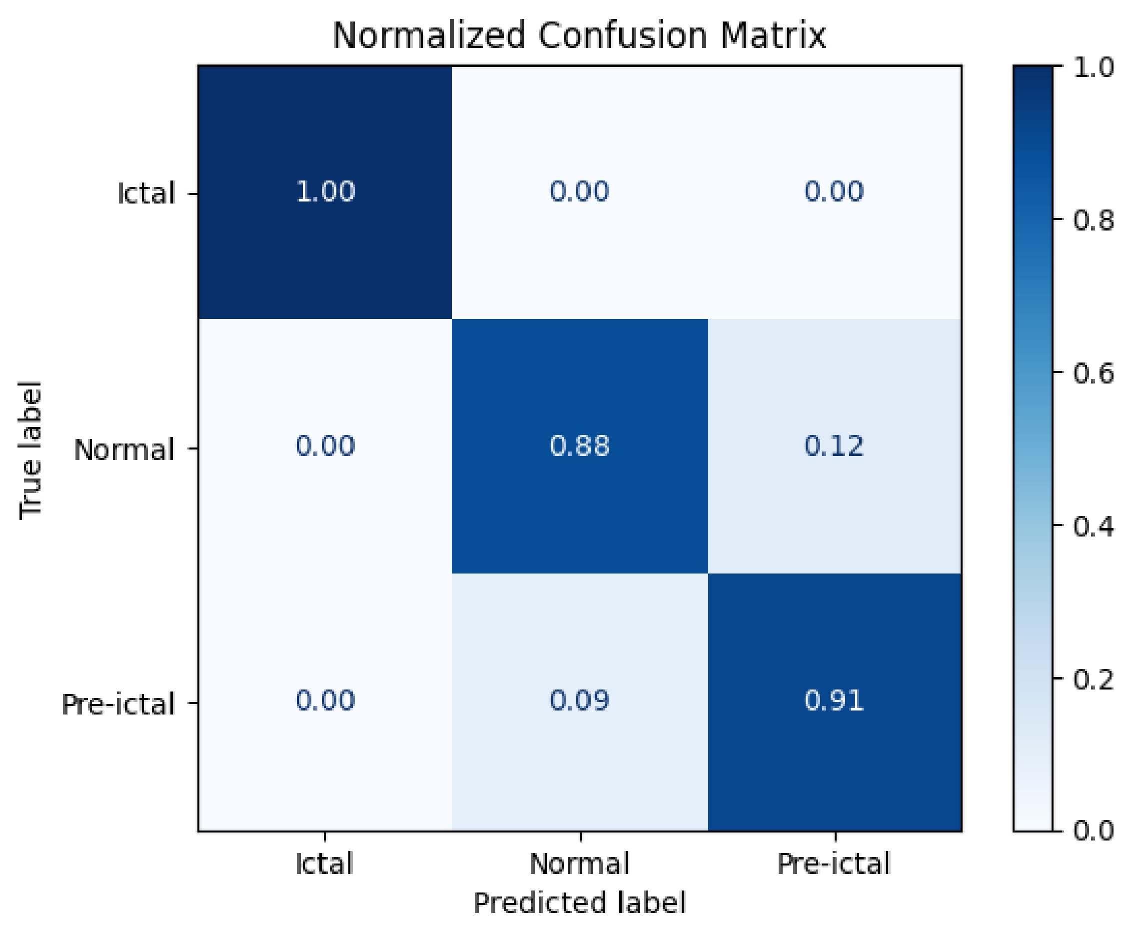
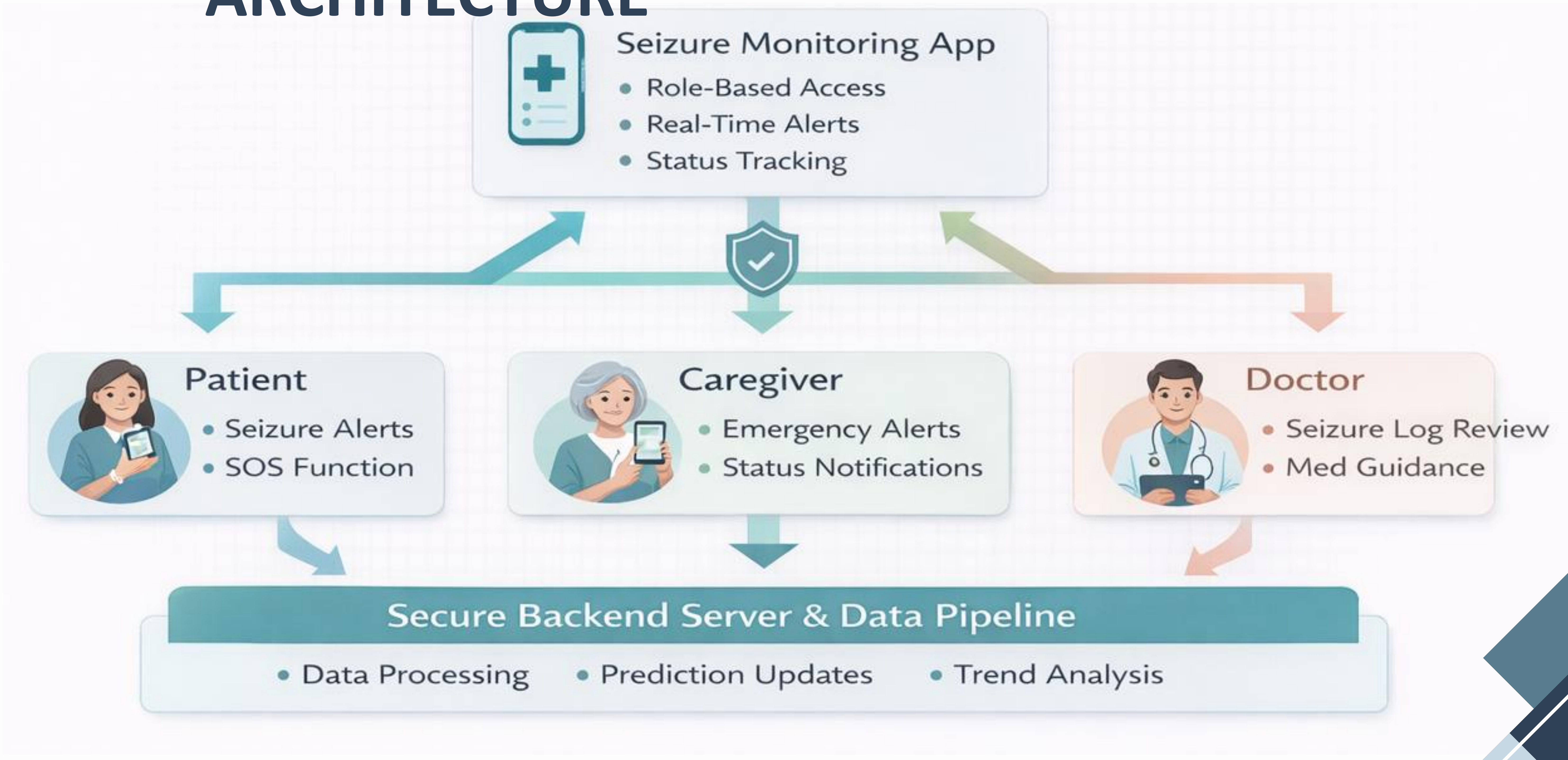


7. MOBILE APPLICATION

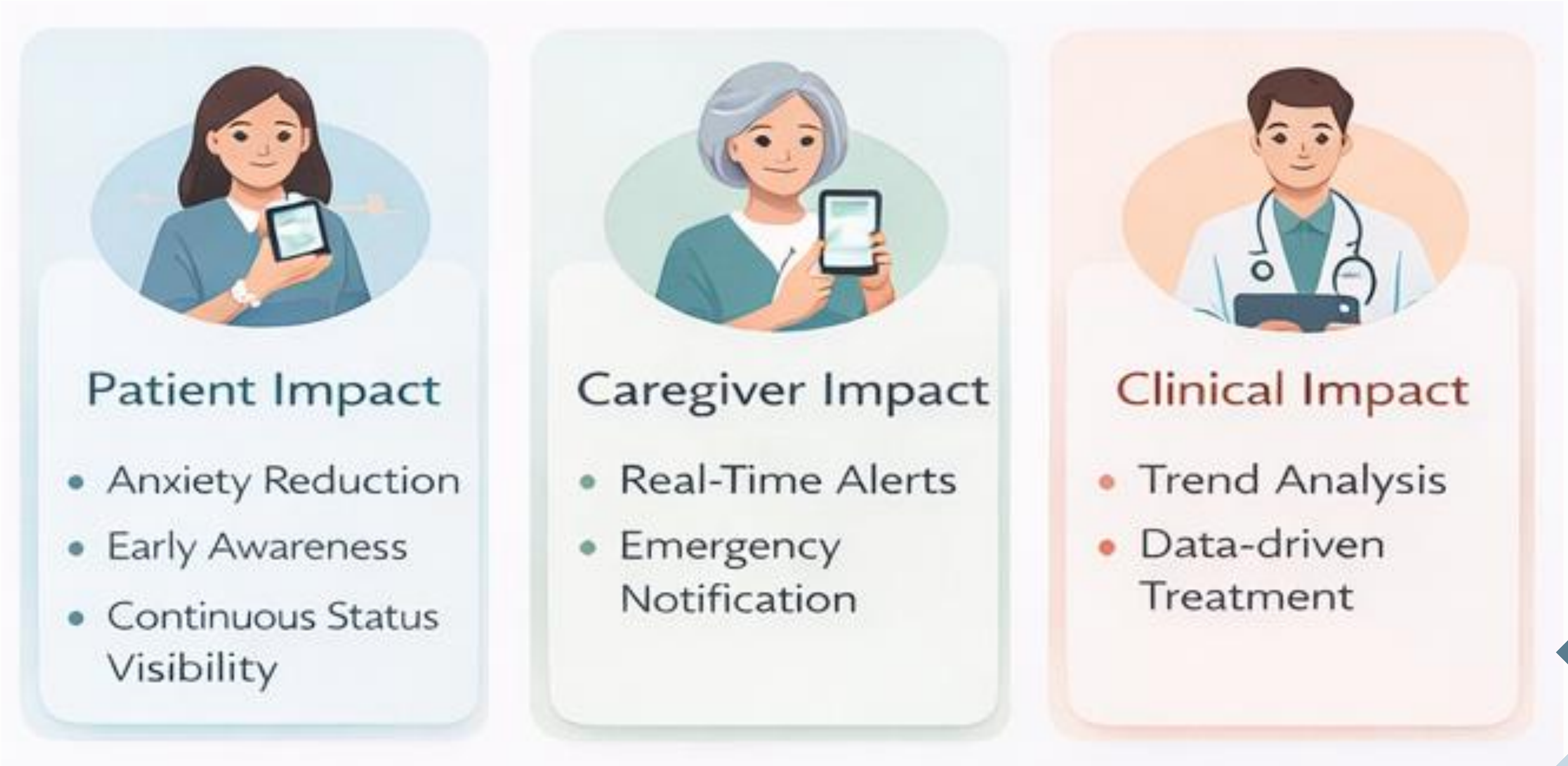
. SYSTEM ARCHITECTURE



. UI/UX LAYER
HUMAN INTERFACE DESIGN

- Translates AI predictions into patient-centered action
- Role-based access (Patient – Caregiver – Doctor)
- Interface layer between prediction and real-life response

. CLINICAL & PSYCHOLOGICAL IMPACT



. EMERGENCY & SAFETY LAYER

8.TIMELINE


Phase 1A — Research & Clinical Foundation
Phase 1B — Baseline Technical
Core Development & System Integration
Phase 3 Optimization
Phase 4 Validation & Finalization

9. NEXT STEP

AI Model Enhancement & Reliability Optimization

- Model refinement & hyperparameter tuning
- False alarm reduction
- Threshold calibration & validation

Objective: Improve clinical reliability and reduce false positives.

Backend & Mobile Application Development

- Backend architecture & secure database
- Real-time API communication
- Mobile application implementation
- AI-powered Clinical Decision Support Chatbot

Objective: Enable secure and intelligent continuous monitoring.

Hardware Prototype Finalization

- Wearable refinement & calibration
- Power optimization
- Stable MCU–Bluetooth communication

Objective: Deliver a reliable wearable prototype.

Full System Integration

- AI inference integration
- Wearable-to-mobile connectivity
- Backend synchronization
- End-to-end testing

Objective: Achieve a fully integrated working system.

Monitoring Expansion

- Additional physiological sensors
- Multi-signal fusion validation

Objective: Enhance monitoring robustness.

Key Partners

- Hospitals and neurology clinics
- Medical researchers and neurologists
- Wearable device hardware suppliers
- IoT cloud service providers
- AI and data analytics platforms

Partner Activities

- Clinical validation and medical guidance
- Supplying wearable sensors and electronic components
- Providing cloud infrastructure and data security
- Supporting research and data analysis

Key Resources from Partners

- Medical expertise and clinical data
- Hardware components and sensors
- Cloud computing and secure data storage

Key Activities

- Physiological signal processing and analysis
- Training and validating AI models
- Developing and maintaining the mobile application
- Integrating wearable devices with IoT platforms
- Monitoring system performance and user feedback

Key Resources

- AI algorithms for seizure prediction
- Wearable device hardware and sensors
- Mobile application platform
- IoT infrastructure for data transmission
- Medical datasets and clinical expertise
- Skilled technical and research team

Value Proportions

- Early seizure prediction before onset
- Continuous monitoring and instant alerts
- Non-invasive and cost-effective solution
- Improved safety during sleep
- Data visualization and medical reports for doctors
- AI chatbot for continuous patient support

Customer Relationships

- Continuous digital support via mobile application
- Automated alerts and notifications
- AI chatbot assistance
- Periodic medical reports and insights

Relationship Cost

- Moderate (mainly software maintenance and cloud services)

Channels

- Hospitals and neurology clinics
- Healthcare partnerships
- Digital health platforms
- Channel Integration
- Wearable device integrated with mobile app
- IoT platform connecting patients, caregivers, and doctors

Customer Segments

- Epilepsy patients (especially drug-resistant epilepsy)
- Caregivers and family members
- Neurologists and hospitals
- Healthcare providers and medical institutions

Cost Structure

- Wearable device development and sensors
- AI model development and training
- Cloud infrastructure and data storage
- Mobile application development and maintenance
- Testing and clinical validation

Revenue Streams

- Sale of wearable devices
- Monthly or annual subscription for mobile app services
- Licensing solutions to hospitals and clinics
- B2B healthcare partnerships

Payment Methods

- Subscription-based payments
- Institutional contracts
- Direct device purchase

Designed For: CREATIVA Tech

Designed by: AURA

Date: 22/1/2026

Version:1.0

Business Model Canvas

10.OUR TEAM

Arwa Mohamed

- Research & Documentation
- Market study
- Hardware design
- Embedded System

Nouran Mohamed

- Project Manager
- Research & Documentation
- AI / Signal Processing
- Design UI/UX

Reem wael

- Research
- Signal Processing
- Feature extraction
- Train AI models

Emad Samuel

- Research
- Market study
- Hardware design
- Mobile App Developer

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- https://openneuro.org/datasets/ds005873/versions/1.1.0

THANK you